Sahil Kumar Sharma

ELECTRONICS & EMBEDDED SYSTEMS STUDENT

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PROFILE

Highly motivated Electronics Engineering student with hands-on expertise in embedded systems, RF communication, PCB design, and IoT architectures. Proven track record of building end-to-end hardware–software solutions—from satellite demonstrators to biomedical devices—with a strong competitive edge in robotics and circuit design. Seeking a research or engineering internship to apply and deepen skills in embedded intelligence and hardware innovation.

EDUCATION

Keshav Mahavidyalaya, University of Delhi 2023 – Present
B.Sc. (Hons.) Electronics — New Delhi

Guru Harkrishna Public School Completed 2021
CBSE Class X & XII — New Delhi

TECHNICAL SKILLS

Languages	C, C++, Python · Java (Intermediate)
Microcontrollers	Arduino (Uno / Nano / Mega), ESP-32, Raspberry Pi, 8085 / 8086 Microprocessors
Design Tools	KiCad, LTspice, Proteus, Arduino IDE, Scilab
Domains	Embedded Systems, PCB Design,Communication, Analog & Digital Electronics, IoT, Signal Processing

PROJECTS

SHAPE-SAT — Student High-Altitude Payload Experiment Satellite

ESP-32 · Python · IoT · Telemetry

- Designed a solar-powered IoT satellite demonstrator replicating real small-satellite architecture with autonomous mission modes and energy-aware power management.
- Integrated multi-sensor payload, real-time telemetry pipeline, and Python-based ground station with live data visualization; deployed onboard Wi-Fi dashboard for in-flight monitoring.

ECG Monitoring Device

Arduino Nano · AD8232 · Python · Matplotlib

- Built a compact real-time ECG acquisition system using the AD8232 biosensor; resolved voltage stability issues with a 3.3V adapter module and visualized filtered waveforms in Python.
- Actively integrating ML-based cardiac anomaly detection and migrating to Raspberry Pi Pico for enhanced signal-processing throughput.

Custom RF Transmitter & Receiver System

Arduino · NRF24L01 · PCB Design

- Engineered a 100 m-range wireless control system using NRF24L01 modules; applied the platform to build a 4×4-drive RoboSoccer car and achieve basic drone stabilization.

RC Plane with Self-Built Transceiver

PCB Fabrication · BLDC · RF Comm · Servo Control

- Designed and fabricated a fully functional RC aircraft—from custom PCB layout and firmware to aerodynamically optimized airframe—achieving stable, responsive flight via self-developed RF link.

Snake Game on WS2812B LED Matrix

Arduino Uno · WS2812B · Linked List · Joystick

- Recreated the classic Snake game on a 16×18 addressable LED matrix using a linked-list data structure; implemented dynamic speed scaling, random food spawning, and score display.

ACHIEVEMENTS

- **1st Place** — Maze Solver Car, DDUC Tech Fest
- **Finalist** — Robot Fight Competition, IIT Delhi
- **2nd Place** — Electromania '25, Shaheed Rajguru College (University of Delhi)

POSTIONS OS RESPONSIBILITY

CESAR — College Robotics Club — Vice President
Keshav Mahavidyalaya, University of Delhi

- Lead cross-departmental robotics initiatives, mentoring 50+ students and organizing inter-college competitions.

ELEKTRONIKA — Departmental Society — Robotics Head
Keshav Mahavidyalaya, University of Delhi

- Drive technical workshops in robotics and embedded systems, elevating department-wide hardware proficiency.